Chapter 2. Cauchy's Theorem and Its Applications

Goursat's theorem, Cauchy's theorem in a disc, Cauchy integral formulas, Morera, Schwarz reflection, Runge

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Overview

Loosely, Cauchy's theorem says that if f is holomorphic in an open set Ω and $\gamma \subset \Omega$ is a closed curve whose interior is also in Ω, then

$$\int_\gamma f(z)\,dz = 0. \tag{1}$$

Defining the "interior" of a general curve is not easy, so we use "toy contours" with obvious interiors. It suffices that f have a primitive (Corollary 3.3), and primitives come from *Goursat's theorem*: $\int_T f(z)\,dz = 0$ for a triangle T with interior in Ω. This leads to the *Cauchy integral formula*

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta \qquad (z \text{ inside } C),$$

and hence to regularity (derivatives of all orders, false for real variables), power series, analytic continuation, Liouville's theorem, the fundamental theorem of algebra and Morera's theorem.

1 Goursat's theorem

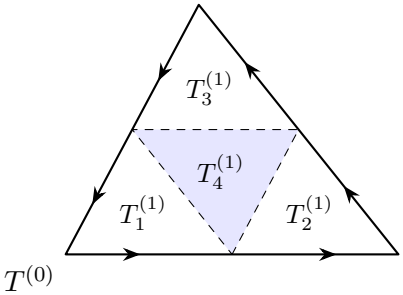


Figure. *Goursat's subdivision: dashed interior edges are traversed twice in opposite directions and cancel, so* $\int_{\partial T^{(0)}} f = \sum_{j=1}^{4} \int_{\partial T_j^{(1)}} f$.

By Corollary 3.3, a primitive forces $\int_\gamma f = 0$ for closed γ; conversely, vanishing integrals over certain curves produce a primitive. Goursat's theorem needs only the existence of f', not its continuity.

Theorem 1.1

If Ω is an open set in $\mathbb{C}$, and $T \subset \Omega$ a triangle whose interior is also contained in Ω, then

$$\int_T f(z)\,dz = 0$$

whenever f is holomorphic in Ω.

Proof. Let $T^{(0)}$ be the positively oriented triangle, with diameter $d^{(0)}$ and perimeter $p^{(0)}$. Joining the midpoints of the sides gives four similar triangles $T_1^{(1)}, \dots, T_4^{(1)}$, consistently oriented; interior sides cancel, so

$$\int_{T^{(0)}} f(z)\,dz = \int_{T_1^{(1)}} f + \int_{T_2^{(1)}} f + \int_{T_3^{(1)}} f + \int_{T_4^{(1)}} f. \tag{2}$$

For some j, $\left|\int_{T^{(0)}} f\right| \le 4\left|\int_{T_j^{(1)}} f\right|$, for otherwise (2) would be contradicted. Call this triangle $T^{(1)}$; its diameter and perimeter are $d^{(1)} = \frac{1}{2}d^{(0)}$, $p^{(1)} = \frac{1}{2}p^{(0)}$. Repeating, we obtain triangles $T^{(0)}, T^{(1)}, \dots, T^{(n)}, \dots$ with

$$\left|\int_{T^{(0)}} f(z)\,dz\right| \le 4^n \left|\int_{T^{(n)}} f(z)\,dz\right|, \qquad d^{(n)} = 2^{-n}d^{(0)}, \quad p^{(n)} = 2^{-n}p^{(0)}.$$

The solid closed triangles $\mathcal{T}^{(n)}$ bounded by $T^{(n)}$ are nested compact sets with diameters $\to 0$, so by Proposition 1.4 they share a unique point z_0. Since f is holomorphic at z_0,

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \psi(z)(z - z_0), \qquad \psi(z) \to 0 \text{ as } z \to z_0.$$

The constant $f(z_0)$ and the linear function $f'(z_0)(z - z_0)$ have primitives, so by Corollary 3.3

$$\int_{T^{(n)}} f(z)\,dz = \int_{T^{(n)}} \psi(z)(z - z_0)\,dz. \tag{3}$$

z_0 is in the solid triangle $\mathcal{T}^{(n)}$ and z on its boundary, so $|z - z_0| \le d^{(n)}$, and by Proposition 3.1(iii)

$$\Big|\int_{T^{(n)}} f(z)\,dz\Big| \le \epsilon_n d^{(n)} p^{(n)} = \epsilon_n 4^{-n} d^{(0)} p^{(0)}, \qquad \epsilon_n = \sup_{z \in T^{(n)}} |\psi(z)| \to 0.$$

Hence $\big|\int_{T^{(0)}} f\big| \le 4^n \big|\int_{T^{(n)}} f\big| \le \epsilon_n d^{(0)} p^{(0)}$, and letting $n \to \infty$ concludes the proof. ∎

Corollary 1.2

If f is holomorphic in an open set Ω that contains a rectangle R and its interior, then $\int_R f(z)\,dz = 0$.

Proof. Split R by a diagonal into two triangles T_1, T_2, oriented consistently; then $\int_R f = \int_{T_1} f + \int_{T_2} f = 0$. ∎

2 Local existence of primitives and Cauchy's theorem in a disc

Motivation. Integrating along polygonal paths turns Goursat's theorem into a primitive on a disc.

Theorem 2.1

A holomorphic function in an open disc has a primitive in that disc.

Proof. Translate so that D is centered at 0. For $z \in D$ let γ_z be the polygonal path from 0 horizontally to $\mathrm{Re}(z)$ and then vertically to z, and define

$$F(z) = \int_{\gamma_z} f(w)\,dw.$$

For $z, z + h \in D$, in $F(z + h) - F(z) = \int_{\gamma_{z+h}} f - \int_{\gamma_z} f$ the common segments cancel; completing a square and a triangle and applying Goursat for triangles and rectangles leaves

$$F(z + h) - F(z) = \int_{\eta} f(w)\,dw,$$

η the segment from z to $z + h$. Since f is continuous at z, $f(w) = f(z) + \psi(w)$ with $\psi(w) \to 0$ as $w \to z$. Hence

$$F(z + h) - F(z) = f(z) \int_{\eta} dw + \int_{\eta} \psi(w)\,dw. \tag{4}$$

The constant 1 has primitive w, so the first integral is h (Theorem 3.2); and $\big|\int_\eta \psi(w)\,dw\big| \le \sup_{w \in \eta} |\psi(w)|\,|h|$. The supremum tends to 0 as $h \to 0$, so (4) gives $\lim_{h \to 0} \frac{F(z+h) - F(z)}{h} = f(z)$. ∎

So holomorphic functions have primitives *locally*; the same holds on "toy contours" below.

Theorem 2.2 (Cauchy's theorem for a disc)

If f is holomorphic in a disc, then $\int_\gamma f(z)\,dz = 0$ for any closed curve γ in that disc.

Proof. f has a primitive (Theorem 2.1); apply Corollary 3.3. ∎

Corollary 2.3

Suppose f is holomorphic in an open set containing the circle C and its interior. Then $\int_C f(z)\,dz = 0$.

Proof. Let D be the disc bounded by C. There is a slightly larger disc $D' \supset \overline{D}$ on which f is holomorphic; apply Cauchy's theorem in D'. ∎

Definition (toy contour)

A **toy contour** is a closed curve with obvious interior for which the construction of Theorem 2.1 works near the curve and its interior; its **positive orientation** keeps the interior on the left.

Circles, triangles and rectangles are toy contours, as is the **keyhole** Γ: a large and a small almost-complete circle joined by a narrow corridor. If f is holomorphic near Γ and its interior, it is holomorphic inside a slightly larger keyhole Γ' with $\overline{\operatorname{int}\Gamma} \subset \operatorname{int}\Gamma'$. Fix $z_0 \in \operatorname{int}\Gamma'$ and join it to z by a path γ_z of horizontal and vertical segments in $\operatorname{int}\Gamma'$; by Corollary 1.2 $F(z) = \int_{\gamma_z} f$ does not depend on the path, F is a primitive, and $\int_\Gamma f = 0$. In general, for a toy contour γ,

$$\int_\gamma f(z)\,dz = 0$$

whenever f is holomorphic in an open set containing γ and its interior. Other examples used later: rectangular and multiple keyholes, semicircle, indented semicircle, sector, parallelogram. Appendix B proves the Jordan curve theorem for piecewise-smooth curves and Cauchy's theorem in that setting.

3 Evaluation of some integrals

We show how some integrals may be evaluated using Cauchy's theorem; a systematic approach via residues is in Chapter 3.

Example 1

If $\xi \in \mathbb{R}$, then

$$e^{-\pi\xi^2} = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x\xi}\,dx, \tag{5}$$

i.e. $e^{-\pi x^2}$ is its own Fourier transform. For $\xi = 0$ this is $\int_{-\infty}^{\infty} e^{-\pi x^2}\,dx = 1$. Let $\xi > 0$, $f(z) = e^{-\pi z^2}$ (entire), and γ_R the positively oriented rectangle with vertices R, $R + i\xi$, $-R + i\xi$, $-R$. By Cauchy's theorem

$$\int_{\gamma_R} f(z)\,dz = 0. \tag{6}$$

The integral over the real segment is $\int_{-R}^{R} e^{-\pi x^2}\,dx \to 1$. The integral over the right vertical side is $I(R) = \int_0^\xi f(R+iy)\,i\,dy = \int_0^\xi e^{-\pi(R^2+2iRy-y^2)}\,i\,dy$, and $|I(R)| \le Ce^{-\pi R^2} \to 0$ (ξ fixed);

similarly for the left side. The top side contributes

$$\int_R^{-R} e^{-\pi(x+i\xi)^2}\,dx = -e^{\pi\xi^2}\int_{-R}^{R} e^{-\pi x^2}e^{-2\pi i x\xi}\,dx.$$

Letting $R\to\infty$, (6) gives $0 = 1 - e^{\pi\xi^2}\int_{-\infty}^{\infty} e^{-\pi x^2}e^{-2\pi i x\xi}\,dx$. If $\xi < 0$ use the symmetric rectangle in the lower half-plane. (This *shifting of the contour* has many applications.)

Example 2

$$\int_0^\infty \frac{1-\cos x}{x^2}\,dx = \frac{\pi}{2}.$$

Integrate $f(z) = (1-e^{iz})/z^2$ over the indented semicircle: $[-R,-\epsilon]$, $[\epsilon,R]$, the negatively oriented γ_ϵ^+ and the positively oriented γ_R^+. By Cauchy's theorem

$$\int_{-R}^{-\epsilon}\frac{1-e^{ix}}{x^2}\,dx + \int_{\gamma_\epsilon^+}\frac{1-e^{iz}}{z^2}\,dz + \int_\epsilon^R \frac{1-e^{ix}}{x^2}\,dx + \int_{\gamma_R^+}\frac{1-e^{iz}}{z^2}\,dz = 0.$$

Since $|(1-e^{iz})/z^2| \le 2/|z|^2$ in the upper half-plane, the integral over γ_R^+ tends to 0 as $R\to\infty$, so $\int_{|x|\ge\epsilon}\frac{1-e^{ix}}{x^2}\,dx = -\int_{\gamma_\epsilon^+}\frac{1-e^{iz}}{z^2}\,dz$. Now $f(z) = \frac{-iz}{z^2} + E(z)$ with E bounded near 0, and on γ_ϵ^+, $z = \epsilon e^{i\theta}$, $dz = i\epsilon e^{i\theta}d\theta$, so

$$\int_{\gamma_\epsilon^+}\frac{1-e^{iz}}{z^2}\,dz \to \int_\pi^0 (-i)(i)\,d\theta = -\pi \quad \text{as } \epsilon\to 0.$$

Taking real parts, $\int_{-\infty}^{\infty}\frac{1-\cos x}{x^2}\,dx = \pi$; the integrand is even.

4 Cauchy's integral formulas

Motivation. Applied to $f(\zeta)/(\zeta - z)$, Cauchy's theorem determines f inside a disc from its boundary values, giving infinite differentiability and Taylor series.

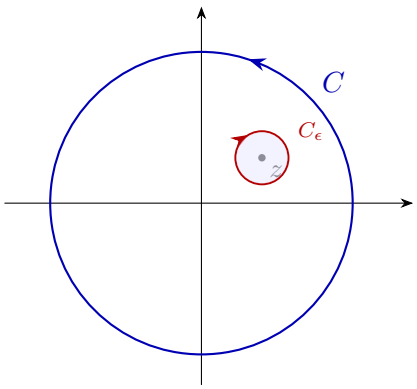


Figure. *$f(\zeta)/(\zeta - z)$ is holomorphic between C and C_ϵ, so both circles give the same integral; as $\epsilon\to 0$ the small one tends to $2\pi i f(z)$.*

Integral representations recover a function on a large set from its values on a smaller one (cf. the Poisson integral in the disc).

Theorem 4.1

Suppose f is holomorphic in an open set containing the closure of a disc D. If

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta \qquad \text{for any point } z \in D.$$

Proof. Fix $z \in D$ and take the keyhole $\Gamma_{\delta,\epsilon}$ omitting z: C and a circle of radius ϵ about z, joined by a corridor of width δ. $F(\zeta) = f(\zeta)/(\zeta - z)$ is holomorphic away from z, so $\int_{\Gamma_{\delta,\epsilon}} F = 0$. As $\delta \to 0$ the corridor integrals cancel by continuity, leaving C (positive) and C_ϵ (negative). Write

$$F(\zeta) = \frac{f(\zeta) - f(z)}{\zeta - z} + \frac{f(z)}{\zeta - z}. \tag{8}$$

The first term is bounded (as f is holomorphic), so its integral over C_ϵ tends to 0 as $\epsilon \to 0$. And

$$\int_{C_\epsilon} \frac{f(z)}{\zeta - z}\,d\zeta = -f(z)\int_0^{2\pi} \frac{\epsilon i e^{-it}}{\epsilon e^{-it}}\,dt = -f(z)2\pi i.$$

So in the limit $0 = \int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta - 2\pi i f(z)$. ∎

Remark

The same holds for other toy contours: for a positively oriented rectangle R, $f(z) = \frac{1}{2\pi i}\int_R \frac{f(\zeta)}{\zeta - z}\,d\zeta$ inside R (rectangular keyhole), and the integral vanishes for z outside R.

Corollary 4.2

If f is holomorphic in an open set Ω, then f has infinitely many complex derivatives in Ω. Moreover, if $C \subset \Omega$ is a circle whose interior is also contained in Ω, then

$$f^{(n)}(z) = \frac{n!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\,d\zeta$$

for all z in the interior of C.

Proof. Induction on n; $n = 0$ is the Cauchy integral formula. Suppose f has up to $n-1$ complex derivatives with $f^{(n-1)}(z) = \frac{(n-1)!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z)^n}\,d\zeta$. For small h,

$$\frac{f^{(n-1)}(z+h) - f^{(n-1)}(z)}{h} = \frac{(n-1)!}{2\pi i}\int_C f(\zeta)\frac{1}{h}\Big[\frac{1}{(\zeta - z - h)^n} - \frac{1}{(\zeta - z)^n}\Big]d\zeta. \tag{9}$$

Since $A^n - B^n = (A-B)[A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}]$, with $A = 1/(\zeta - z - h)$, $B = 1/(\zeta - z)$ the bracket in (9) is

$$\frac{h}{(\zeta - z - h)(\zeta - z)}[A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}].$$

For small h, $z + h$ and z stay at a finite distance from C, so as $h \to 0$ the quotient converges to

$$\frac{(n-1)!}{2\pi i}\int_C f(\zeta)\frac{1}{(\zeta - z)^2}\frac{n}{(\zeta - z)^{n-1}}\,d\zeta = \frac{n!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\,d\zeta.$$

∎

The formulas of Theorem 4.1 and Corollary 4.2 are the **Cauchy integral formulas**.

Corollary 4.3 (Cauchy inequalities)

If f is holomorphic in an open set that contains the closure of a disc D centered at z_0 and of radius R, then

$$|f^{(n)}(z_0)| \leq \frac{n!\,\|f\|_C}{R^n},$$

where $\|f\|_C = \sup_{z\in C}|f(z)|$ denotes the supremum of $|f|$ on the boundary circle C.

Proof. By the Cauchy integral formula,

$$|f^{(n)}(z_0)| = \Big|\frac{n!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\,d\zeta\Big| = \frac{n!}{2\pi}\Big|\int_0^{2\pi} \frac{f(z_0 + Re^{i\theta})}{(Re^{i\theta})^{n+1}} Rie^{i\theta}\,d\theta\Big| \leq \frac{n!}{2\pi}\frac{\|f\|_C}{R^n}2\pi.$$

■

Theorem 4.4

Suppose f is holomorphic in an open set Ω. If D is a disc centered at z_0 and whose closure is contained in Ω, then f has a power series expansion at z_0

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

for all $z \in D$, and the coefficients are given by $a_n = \dfrac{f^{(n)}(z_0)}{n!}$ for all $n \geq 0$.

Proof. Fix $z \in D$. By the Cauchy integral formula

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta, \tag{10}$$

C the boundary of D. Write

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0 - (z - z_0)} = \frac{1}{\zeta - z_0}\cdot\frac{1}{1 - \left(\frac{z - z_0}{\zeta - z_0}\right)}. \tag{11}$$

Since $\zeta \in C$ and $z \in D$ is fixed, there is $0 < r < 1$ with $\left|\frac{z-z_0}{\zeta - z_0}\right| < r$, so

$$\frac{1}{1 - \left(\frac{z - z_0}{\zeta - z_0}\right)} = \sum_{n=0}^{\infty}\left(\frac{z - z_0}{\zeta - z_0}\right)^n, \tag{12}$$

uniformly for $\zeta \in C$. Interchanging sum and integral in (10)–(12),

$$f(z) = \sum_{n=0}^{\infty}\Big(\frac{1}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\,d\zeta\Big)(z - z_0)^n.$$

The formula for a_n follows from the Cauchy integral formulas for derivatives (or by differentiating the series). ■

This again shows infinite differentiability. The expansion converges in *any* disc centered at z_0 with closure in Ω; an entire function's expansion about 0 converges on all of $\mathbb{C}$.

Corollary 4.5 (Liouville's theorem)

If f is entire and bounded, then f is constant.

Proof. It suffices to show $f' = 0$ ($\mathbb{C}$ is connected; Corollary 3.4). For each $z_0 \in \mathbb{C}$ and $R > 0$ the Cauchy inequalities give $|f'(z_0)| \leq B/R$, B a bound for f. Let $R \to \infty$. ■

Corollary 4.6

Every non-constant polynomial $P(z) = a_n z^n + \cdots + a_0$ with complex coefficients has a root in $\mathbb{C}$.

Proof. If P has no roots, $1/P$ is a bounded entire function: with $a_n \neq 0$, $\frac{P(z)}{z^n} = a_n + (\frac{a_{n-1}}{z} + \cdots + \frac{a_0}{z^n})$, the bracket $\to 0$, so $|P(z)| \geq c|z|^n$ for $|z| > R$ ($c = |a_n|/2$), and P is continuous without roots on $|z| \leq R$, hence bounded below there. Liouville makes $1/P$ constant, a contradiction. ■

Corollary 4.7

Every polynomial $P(z) = a_n z^n + \cdots + a_0$ of degree $n \geq 1$ has precisely n roots in $\mathbb{C}$. If these roots are denoted by $w_1, \ldots, w_n$, then P can be factored as $P(z) = a_n(z - w_1)(z - w_2)\cdots(z - w_n)$.

Proof. P has a root w_1. Writing $z = (z - w_1) + w_1$ and expanding, $P(z) = b_n(z - w_1)^n + \cdots + b_1(z - w_1) + b_0$ with $b_n = a_n$; $P(w_1) = 0$ forces $b_0 = 0$, so $P(z) = (z - w_1)Q(z)$ with Q of degree $n - 1$. By induction $P(z) = c(z - w_1)\cdots(z - w_n)$, and comparing coefficients of z^n, $c = a_n$. ■

Finally, *analytic continuation*: a holomorphic function is determined by its values on arbitrarily small sets (Ω connected).

Theorem 4.8

Suppose f is a holomorphic function in a region Ω that vanishes on a sequence of distinct points with a limit point in Ω. Then f is identically 0.

Proof. Let $z_0 \in \Omega$ be a limit point of zeros w_k. First, $f \equiv 0$ near z_0: on a disc $D \subset \Omega$ about z_0, $f(z) = \sum a_n(z - z_0)^n$; otherwise with m the least index with $a_m \neq 0$, $f(z) = a_m(z - z_0)^m(1 + g(z - z_0))$, $g \to 0$, and $f(w_k) \neq 0$ for $w_k \neq z_0$ close to z_0, a contradiction.

Let U be the interior of $\{f = 0\}$, open and non-empty. U is closed in Ω: if $U \ni z_n \to z \in \Omega$, then $f(z) = 0$ and f vanishes near z by the first part. So $\Omega = U \cup (\Omega - U)$ with both open, and $U = \Omega$ by connectedness. ■

Corollary 4.9

If f, g are holomorphic in a region Ω and $f = g$ on a non-empty open subset (or on a sequence of distinct points with limit point in Ω), then $f = g$ on Ω.

If f, F are analytic in regions $\Omega \subset \Omega'$ and agree on Ω, F is an **analytic continuation** of f into Ω'; by the corollary it is unique.

5 Further applications

Motivation. Morera's converse, uniform limits, parameter integrals, reflection and polynomial approximation all follow from the integral formula.

5.1 Morera's theorem

Theorem 5.1 (Morera)

Suppose f is a continuous function in the open disc D such that for any triangle T contained in D, $\int_T f(z)\,dz = 0$. Then f is holomorphic.

Proof. By the proof of Theorem 2.1, f has a primitive F in D with $F' = f$. By regularity (Corollary 4.2), F is indefinitely (hence twice) complex differentiable, so f is holomorphic. ■

5.2 Sequences of holomorphic functions

Theorem 5.2

If $\{f_n\}_{n=1}^{\infty}$ is a sequence of holomorphic functions that converges uniformly to a function f in every compact subset of Ω, then f is holomorphic in Ω.

Proof. For a disc D, $\overline{D} \subset \Omega$, and a triangle $T \subset D$, $\int_T f_n = 0$ (Goursat). Uniform convergence makes f continuous and $\int_T f = \lim \int_T f_n = 0$, so f is holomorphic in D by Morera. ■

False for real variables: every continuous function on $[0,1]$ is a uniform limit of polynomials (Weierstrass).

Theorem 5.3

Under the hypotheses of the previous theorem, the sequence of derivatives $\{f_n'\}_{n=1}^{\infty}$ converges uniformly to f' on every compact subset of Ω.

Proof. Assume uniform convergence on Ω. Let $\Omega_\delta = \{z \in \Omega : \overline{D_\delta(z)} \subset \Omega\}$; it suffices that $f_n' \to f'$ uniformly on each Ω_δ, which follows by applying to $F = f_n - f$ the inequality

$$\sup_{z\in\Omega_\delta} |F'(z)| \le \frac{1}{\delta} \sup_{\zeta\in\Omega} |F(\zeta)|, \tag{13}$$

valid for F holomorphic in Ω. For $z \in \Omega_\delta$, $\overline{D_\delta(z)} \subset \Omega$ and by the Cauchy integral formula

$$|F'(z)| = \Big|\frac{1}{2\pi i}\int_{C_\delta(z)} \frac{F(\zeta)}{(\zeta - z)^2}\,d\zeta\Big| \le \frac{1}{2\pi} \sup_{\zeta\in\Omega}|F(\zeta)| \frac{1}{\delta^2} 2\pi\delta = \frac{1}{\delta}\sup_{\zeta\in\Omega}|F(\zeta)|.$$

■

Likewise $f_n^{(k)} \to f^{(k)}$ uniformly on compacta for every k. Theorem 5.2 constructs holomorphic functions as series $F(z) = \sum_{n=1}^{\infty} f_n(z)$ (14) converging uniformly on compacta, e.g. the zeta function (Chapter 6).

5.3 Holomorphic functions defined in terms of integrals

After a linear change of variables we take the interval $[0,1]$.

Theorem 5.4

Let $F(z,s)$ be defined for $(z,s) \in \Omega \times [0,1]$ where Ω is an open set in $\mathbb{C}$. Suppose F satisfies the following properties:
(i) $F(z,s)$ is holomorphic in z for each s.

(ii) F is continuous on $\Omega \times [0,1]$. Then the function f defined on Ω by $f(z) = \int_0^1 F(z,s)\,ds$ is holomorphic.

Proof. Use Riemann sums $f_n(z) = \frac{1}{n}\sum_{k=1}^n F(z,k/n)$, holomorphic by (i). F is uniformly continuous on $\overline{D}\times[0,1]$ ($\overline{D} \subset \Omega$), so for $\epsilon > 0$ there is $\delta > 0$ with $\sup_{z\in\overline{D}}|F(z,s_1)-F(z,s_2)| < \epsilon$ for $|s_1-s_2| < \delta$. For $n > 1/\delta$ and $z \in D$,

$$|f_n(z) - f(z)| = \Big|\sum_{k=1}^n \int_{(k-1)/n}^{k/n} [F(z,k/n) - F(z,s)]\,ds\Big| \le \sum_{k=1}^n \frac{\epsilon}{n} = \epsilon.$$

So $f_n \to f$ uniformly on D, and f is holomorphic in D by Theorem 5.2; hence in Ω. ∎

5.4 Schwarz reflection principle

Holomorphic functions are rigid (some cannot be continued beyond a disc; vanishing on a segment forces $f \equiv 0$), yet they extend by reflection. Let Ω be open and symmetric about $\mathbb{R}$, $\Omega^{\pm}$ its parts in the upper and lower half-planes, $I = \Omega \cap \mathbb{R}$; $\Omega = \Omega^+ \cup I \cup \Omega^-$.

Theorem 5.5 (Symmetry principle)

If f^+ and f^- are holomorphic functions in Ω^+ and Ω^- respectively, that extend continuously to I and $f^+(x) = f^-(x)$ for all $x \in I$, then the function f defined on Ω by

$$f(z) = \begin{cases} f^+(z) & z \in \Omega^+, \\ f^+(z) = f^-(z) & z \in I, \\ f^-(z) & z \in \Omega^- \end{cases}$$

is holomorphic on all of Ω.

Proof. f is continuous; holomorphy at I follows from Morera on a disc $D \subset \Omega$ centered on I. If a triangle $T \subset D$ misses I, $\int_T f = 0$. If an edge or vertex of T lies on I and the rest in the upper half-disc, raise it by ϵ to get T_ϵ with $\int_{T_\epsilon} f = 0$, and let $\epsilon \to 0$. If the interior of T meets I, split T into triangles with an edge or vertex on I. ∎

Theorem 5.6 (Schwarz reflection principle)

If f is holomorphic in Ω^+, extends continuously to I and is real on I, then there is F holomorphic in Ω with $F = f$ on Ω^+.

Proof. Set $F(z) = \overline{f(\overline{z})}$ on Ω^-. Near $\overline{z_0} \in \Omega^+$, $f(\overline{z}) = \sum a_n(\overline{z} - \overline{z_0})^n$, so $F(z) = \sum \overline{a_n}(z - z_0)^n$ is holomorphic in Ω^-. As f is real on I, F extends continuously to I and agrees with f; apply the symmetry principle. ∎

5.5 Runge's approximation theorem

Which compact $K \subset \mathbb{C}$ allow every function holomorphic near K to be a uniform limit of polynomials on K? Discs do (power series). $K = C$ does not: $\int_C 1/z = 2\pi i$ but $\int_C p = 0$ for polynomials p. The right condition is that K^c be connected (necessity by a variant of this example).

Theorem 5.7 (Runge)

Any function holomorphic in a neighborhood of a compact set K can be approximated uniformly on K by rational functions whose singularities are in K^c. If K^c is connected, any function holomorphic in a neighborhood of K can be approximated uniformly on K by polynomials.

The second part follows by "pushing the singularities to infinity". The key is a representation from the Cauchy formula for squares:

Lemma 5.8

Suppose f is holomorphic in an open set Ω, and $K \subset \Omega$ is compact. Then there exist finitely many segments $\gamma_1, \dots, \gamma_N$ in $\Omega - K$ such that

$$f(z) = \sum_{n=1}^{N} \frac{1}{2\pi i} \int_{\gamma_n} \frac{f(\zeta)}{\zeta - z}\, d\zeta \qquad \text{for all } z \in K. \tag{15}$$

Proof. Let $d = c \cdot d(K, \Omega^c)$, $c < 1/\sqrt{2}$, and take a grid of closed axis-parallel squares of side d. Let $\mathcal{Q} = \{Q_1, \dots, Q_M\}$ be those meeting K, with positively oriented boundaries ∂Q_m, and $\gamma_1, \dots, \gamma_N$ the sides not shared by two squares of $\mathcal{Q}$. By the choice of d, each γ_n lies in Ω and misses K (else it would be shared). For $z \in K$ off the square boundaries, $z \in Q_j$ for some j, and by Cauchy's theorem and formula

$$\frac{1}{2\pi i} \int_{\partial Q_m} \frac{f(\zeta)}{\zeta - z}\, d\zeta = \begin{cases} f(z) & m = j, \\ 0 & m \neq j, \end{cases} \qquad \text{so} \qquad f(z) = \sum_{m=1}^{M} \frac{1}{2\pi i} \int_{\partial Q_m} \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

Shared sides are traversed once each way and cancel, giving (15) for such z; as the γ_n lie in K^c, continuity extends (15) to all of K. ∎

Lemma 5.9

For any segment $\gamma \subset \Omega - K$, $\int_\gamma f(\zeta)/(\zeta - z)\, d\zeta$ is a uniform limit on K of rational functions with singularities on γ.

Proof. $\int_\gamma \frac{f(\zeta)}{\zeta - z}\, d\zeta = \int_0^1 \frac{f(\gamma(t))\gamma'(t)}{\gamma(t) - z}\, dt$; as γ misses K, the integrand is uniformly continuous on $K \times [0,1]$, and as in Theorem 5.4 its Riemann sums, rational functions with singularities on γ, converge uniformly on K. ∎

Lemmas 5.8 and 5.9 give the first part of Theorem 5.7. A rational function singular only at z_0 is a polynomial in $1/(z - z_0)$, so the second part follows from:

Lemma 5.10

If K^c is connected and $z_0 \notin K$, then the function $1/(z - z_0)$ can be approximated uniformly on K by polynomials.

Proof. Choose z_1 outside a large open disc D centered at 0 containing K. Then

$$\frac{1}{z - z_1} = -\frac{1}{z_1} \frac{1}{1 - z/z_1} = -\sum_{n=0}^{\infty} \frac{z^n}{z_1^{n+1}},$$

uniformly on K, so $1/(z - z_1)$ and its powers are uniform limits of polynomials on K. It remains to approximate $1/(z - z_0)$ uniformly on K by polynomials in $1/(z - z_1)$. As K^c is connected, there is

a curve $\gamma \subset K^c$ from z_0 to z_1; let $\rho = \frac{1}{2}d(K,\gamma) > 0$ and choose $w_0 = z_0, w_1, \ldots, w_\ell = z_1$ on γ with $|w_j - w_{j+1}| < \rho$.

Claim: if $w, w' \in \gamma$ with $|w - w'| < \rho$, then $1/(z-w)$ is a uniform limit on K of polynomials in $1/(z-w')$. Indeed

$$\frac{1}{z-w} = \frac{1}{z-w'} \cdot \frac{1}{1 - \frac{w-w'}{z-w'}} = \sum_{n=0}^{\infty} \frac{(w-w')^n}{(z-w')^{n+1}},$$

converging uniformly on K, as $|z - w'| \geq 2\rho$ gives $|w - w'|/|z - w'| < 1/2$. Applying the claim successively, $1/(z-w_0)$ is a uniform limit of polynomials in $1/(z-w_1)$, hence in $1/(z-w_2)$, ..., hence in $1/(z-w_\ell) = 1/(z-z_1)$. ■